Interconnect train↔infer workload taxonomy

Research lens: classify **training vs inference** from GPU fabric / collective **bytes + shape** (not NVML util). 24 characterized families across 6 axes — controls, FN (quiet train), FP (loud infer). Sources: ICTD scaffolds, SPAR red-team backlog, Reveal ib0 separation, SemiAnalysis serving regimes.

24 families	**14 / 9 / 1** train / infer / non-ML	**7 FN** quiet train	**4 FP** loud infer	**Hypothesis:** cross-node byte volume and/or collective shape separates LLM train from infer — only if corpus covers low-comm train *and* production-loud serving. Volume-only DDP vs DP-infer is a toy problem.

Six characterization axes

1. Parallelism topology DP · TP · PP · EP · FSDP/ZeRO · 3D mix. Drives collective type and communication graph.	**2. Collective op mix** AllReduce vs AG/RS vs AllToAll vs P2P. Op ratios beat mean bandwidth.
3. Volume × duty cycle Bytes/s × how often. DiLoCo/grad-accum crush duty; heavy TP/MoE inflate volume.	**4. Temporal / phase regime** Train: fwd/bwd/opt. Infer: prefill · midfill · decode · agentic. Disagg = KV handoff spikes.
5. Graph shape Clique (DP) · ring (TP) · chain (PP) · bipartite (disagg) · expert A2A.	**6. Adversarial role** Controls · FN quiet train · FP loud infer. LOO-by-family on both directions.

All 24 families

ID	Workload family	Role	Label	Dominant collective	Vol	Duty / graph	Why it belongs
W01	Dense DP / DDP train	train+	train	AllReduce (grad)	high	every step, periodic · all-to-all clique	Positive control — classic |θ| sync fingerprint
W02	FSDP / ZeRO-3 train	train+	train	AG + RS (param/grad)	sat	per layer / per step · all-to-all clique	Sharded train — high volume, different op mix than DDP
W03	TP + DP train (Megatron-style)	train+	train	AR (TP) + AR (DP)	sat	intra-layer + step · TP ring + DP clique	Multi-collective train baseline for op-mix features
W04	Pipeline 1F1B train	confuse	train	P2P activations	med	bubble + microbatch · pipeline chain	Train without DP AllReduce — volume-only detectors fail
W05	MoE EP train (AllToAll)	train+	train	A2A dispatch/combine	sat	per layer, token-routed · expert all-to-all	Train A2A — same fabric signature as MoE infer (hard case)
W06	DP / replicate infer	infer–	infer	near-none (or rare)	low	request-driven · isolated replicas	Negative control — quietest multi-GPU infer
W07	TP infer — prefill-heavy	infer–	infer	AR per layer (acts)	med	burst then quiet · TP ring	Compute-heavy fill; moderate sustained AR
W08	TP infer — decode-heavy	infer–	infer	AR per token/layer	med	steady low-amp · TP ring	Long-running small messaging — not |θ| spikes
W09	TP infer — midfill / agentic	infer–	infer	AR + KV touch	med	intermittent bursts · TP ring	Continuation fill — between prefill and decode
W10	Disagg prefill↔decode KV	FP	infer	P2P / RDMA KV blobs	high	handoff spikes (GB-scale) · bipartite pools	Primary FP weapon — looks like big training transfers
W11	Heavy continuous-batch TP serve	FP	infer	AR saturated	high	near-continuous · TP ring @ high QPS	Production serving can push bytes into train band
W12	MoE EP infer (AllToAll)	FP	infer	A2A per layer	sat	continuous under load · expert all-to-all	Infer as loud as train on fabric — volume detectors die

All 24 families (continued)

ID	Workload family	Role	Label	Dominant collective	Vol	Duty / graph	Why it belongs
W13	DiLoCo / local SGD (H■1)	FN	train	rare outer AR	low	1/H of steps · clique, sparse in time	Canonical FN — ~500× less sync than DDP
W14	Streaming DiLoCo	FN	train	chunked outer sync	low	fragmented peaks crushed · clique + overlap	Kills peak-bandwidth / periodicity features
W15	Grad-accum / rare sync	FN	train	AR every N microsteps	med	1/N · clique, thinned	Same topology, lower duty — window truth needed
W16	LoRA / PEFT-only sync	FN	train	AR adapters only	low	every step but tiny · clique	MB-scale payload — interconnect twin of Rahman WB-L
W17	PowerSGD / Top-k sparse	FN	train	compressed AR	low	every step, small · clique	Payload ■ |θ| without changing schedule
W18	KV-shaped gradient disguise	FN	train	P2P / chunked like KV	med	engineered to match disagg · bipartite-looking	Train comm forged to look like serving KV
W19	Hybrid duty-cycle (mostly decode)	FN	train	decode AR + rare |θ|	med	infer-shaped + spikes · mixed	Dilution on the wire — STEP/window labels mandatory
W20	Fake collectives on serving	FP	infer	padded AR / zeros	high	injected periodicity · clique-like noise	Cheap FP — breaks volume + period heuristics
W21	Trace-replay serving (Azure/BurstGPT)	infer–	infer	as deployed config	med	real arrival process · production-like	Non-synthetic load shape for adversarial FPR
W22	Fine-tune / short SFT	confuse	train	AR but short run	med	train-like, brief · clique	Reveal: fine-tune ib0≈19 MB/s — duration/regime matter
W23	Intra-node-only train (NVLink, IB quiet)	FN	train	NVLink AR only	near-0	cross-node silent · intra-node	Blinds IB-only monitor — needs NVML fallback layer
W24	Idle / non-ML GPU	neg	non-ML	none	near-0	none · none	Floor baseline + multi-tenant noise floor

Suggested 2-node collection order

Phase	IDs	Notes
P0 smoke (2-node)	W01, W06, W07/W08, W24	Calibrate NCCL + IB vs labels
P0 FN core	W13, W15, W19	DiLoCo H-sweep + rare sync + hybrid
P1 FP core	W10, W11, W12, W20	Disagg KV · heavy TP · MoE EP · fake AR
P1 FN harder	W14, W16, W17, W18	Streaming · LoRA · PowerSGD · KV-forge
P2 shape / meta	W04, W05, W03, W21–W23	PP · EP train · 3D · traces · NVLink-only

Parsa ~14 core → this grid

Typical ICTD/SPAR set (~14): (1) DDP train (2) FSDP train (3) DP infer (4) TP prefill (5) TP decode (6) DiLoCo (7) grad-accum (8) LoRA sync (9) KV disguise (10) hybrid duty (11) disagg KV (12) MoE EP synth/real (13) fake AR on serve (14) idle/non-ML. Expanded to 24 by splitting midfill, streaming DiLoCo, PowerSGD, PP, EP-train, heavy serve, trace-replay, fine-tune regime, and NVLink-only blind spot.

Do not inflate the count with: Idle × N models, AMP vs FP32 on the same DDP recipe, or CIFAR vs WikiText with identical collective schedule — those are NVML/compute variants, not new interconnect classes. Count a family when parallelism, sync policy, collective mix, or serving phase changes the fabric signature.